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Roll No.....

EC-502 (GS)

B.E. V Semester Examination, June 2020

Grading System (GS)

Control Systems

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Reduce the block diagram, shown in figure 1, into a form having one block in the forward path and one in feedback path.

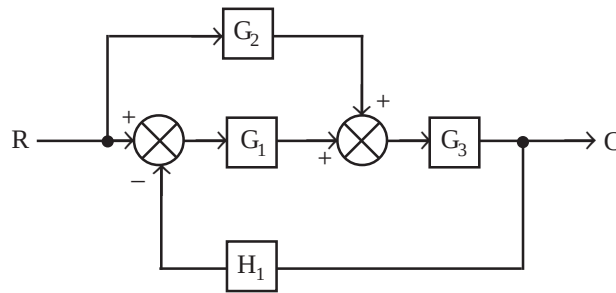


Figure 1

2. Draw the signal flow graph for the following equations.

i) $\frac{d^2y}{dt^2} + \frac{10dy}{dt} - 5y = 3x$

ii) $\frac{d^2y}{dt^2} + 3y = x$

3. Explain the meaning of term 'BIBO' use Routh Hurwitz criterion and test the stability for a system having a characteristics equation.

$$s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 6 = 0$$

4. What do you understand by digital control system why Z-transform is used in digital control system?

5. The open loop transfer function of unity feedback system is given by $G(s) = \frac{50}{(1+0.15s)(s+10)}$

Determine the static error coefficients K_p , K_v and K_a .

[2]

6. The state equation of a linear time invariant system is given below

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t) ; t > 0$$

Find the controllability of the system and calculate the state transition matrix.

7. Write short note on proportional control action and integral control action.

OR

Check the observability of the below given system

$$\dot{x}_1 = x_2, \quad \dot{x}_2 = -2x_1 - 3x_2 + 4 \quad \text{and} \quad y = x_1 + x_2$$

8. Answer any Two of the following.

- Explain the relationship between damping ratio and % overshoot.
- Derive the expression for resonant frequency
- Define state, state variables and state vector
- What are the advantages of state space techniques?
